Regression in machine learning

Last Updated : 19 Jan, 2026

Regression in machine learning is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target). It helps understand how changes in one or more factors influence a measurable outcome and is widely used in forecasting, risk analysis, decision-making and trend estimation.

- Works with real-valued output variables
- Helps to identify strengths and the type of relationships
- Supports both simple and complex predictive models.
- Used for tasks like price prediction, trend forecasting and risk scoring.

Types of Regression

Regression can be classified into different types based on the number of predictor variables and the nature of the relationship between variables:

1. Simple Linear Regression

Simple Linear Regression models the relationship between one independent variable and a continuous dependent variable by fitting a straight line that minimizes the sum of squared errors. It assumes a constant rate of change, meaning the output varies proportionally with the input.

- **Application:** Estimating house price from only its size
- **Advantage:** Highly interpretable due to its simple mathematical structure
- **Disadvantage:** Cannot capture curved or complex data patterns

2. Multiple Linear Regression

Multiple Linear Regression extends simple linear regression by incorporating multiple independent variables to predict a continuous outcome. Each predictor is assigned a coefficient that reflects its individual impact while holding other variables constant.

- **Application:** Predicting house prices using multiple factors like size, location, age and number of rooms
- **Advantage:** Captures the combined influence of many factors simultaneously
- **Disadvantage:** Performance drops in the presence of multicollinearity (features highly correlated with each other)

3. Polynomial Regression

Polynomial Regression models non-linear relationships by transforming input features into higher-degree polynomial terms (e.g x^2, x^3). Although it fits non-linear curves it remains a linear model in terms of parameters.

- **Application:** Modelling curved growth trends like population increase or temperature variation
- **Advantage:** Effectively captures non-linear relationships without switching to non-linear algorithms
- **Disadvantage:** Higher-degree polynomials may lead to overfitting and unstable predictions

4. Ridge and Lasso Regression

Ridge and Lasso are regularized linear regression techniques that add penalty terms to limit large coefficients and reduce overfitting. Ridge (L2) shrinks coefficients smoothly, while Lasso (L1) can reduce some coefficients to zero, enabling feature selection.

- **Application:** Used in high-dimensional datasets like marketing attribution or gene expression data
- **Advantage:** Controls overfitting and improves generalization, especially with many predictors
- **Disadvantage:** Penalty terms make model interpretation less straightforward

5. Support Vector Regression (SVR)

Support Vector Regression applies the principles of Support Vector Machines to regression tasks. It fits a function within a defined margin (epsilon-tube) and penalizes errors only when predictions fall outside this boundary. Kernel functions allow SVR to model non-linear relationships.

- **Application:** Predicting continuous outcomes such as stock values or energy consumption
- **Advantage:** Works well with high-dimensional, complex datasets and non-linear patterns
- **Disadvantage:** Computationally intensive and requires careful tuning of kernels and parameters

6. Decision Tree Regression

Decision Tree Regression splits the data into hierarchical branches based on feature thresholds.. Each internal node represents a decision question and leaf nodes represent predicted continuous values. It learns patterns by recursively partitioning the data to minimize prediction errors.

- **Application:** Predicting customer spending behavior based on demographic and financial features
- **Advantage:** Easy to visualize and understand decision logic
- **Disadvantage:** Easily overfits, especially when the tree becomes deep and complex

7. Random Forest Regression

Random Forest Regression is an ensemble method that builds multiple decision trees using different data samples and averages their predictions. This reduces the overfitting tendency of single trees and improves accuracy through diversity (bagging). Each tree captures a slightly different aspect of the data.

- **Application:** Sales forecasting, demand planning, churn prediction
- **Advantage:** High accuracy and robust performance even on noisy datasets

- **Disadvantage:** Acts as a black-box model, making interpretation difficult due to many trees

Regression Evaluation Metrics

Evaluation in machine learning measures the performance of a model. Here are some popular evaluation metrics for regression:

- **Mean Absolute Error (MAE):** The average absolute difference between the predicted and actual values of the target variable.
- **Mean Squared Error (MSE):** The average squared difference between the predicted and actual values of the target variable.
- **Root Mean Squared Error (RMSE):** Square root of the mean squared error.
- **Huber Loss:** A hybrid loss function that transitions from MAE to MSE for larger errors, providing balance between robustness and MSE's sensitivity to outliers.
- **R2 – Score:** Higher values indicate better fit ranging from 0 to 1.

Implementing Linear Regression in Python

Here we apply linear regression to a housing dataset to predict house prices. The following Python code demonstrates how this model is implemented.

You can download dataset from here.

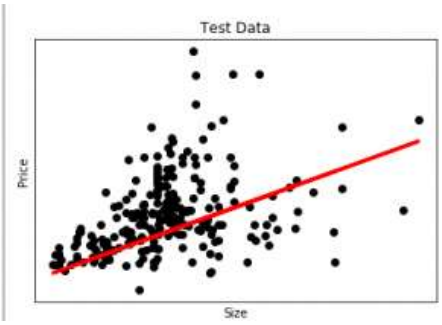
- The housing dataset is loaded with lot size as the input feature and price as the target variable.
- The data is split into training and testing sets and a linear regression model is trained.
- Predictions on test data are plotted against actual values.

```
import pandas as pd
from sklearn import linear_model
import numpy as np
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')
df = pd.read_csv("Housing.csv")
Y = df['price']
X = df['lotsize']
X = X.to_numpy().reshape(len(X), 1)
Y = Y.to_numpy().reshape(len(Y), 1)

X_train = X[:-250]
X_test = X[-250:]
Y_train = Y[:-250]
Y_test = Y[-250:]
plt.scatter(X_test, Y_test, color='black')
plt.title('Test Data')
plt.xlabel('Size')
plt.ylabel('Price')
plt.xticks(())
plt.yticks(())
regr = linear_model.LinearRegression()
regr.fit(X_train, Y_train)

plt.plot(X_test, regr.predict(X_test), linewidth=3, color='red')
plt.savefig("regression_plot.png")
print("Plot saved as regression_plot.png")
```

Output:



Here in this graph we plot the test data. The red line indicates the best fit line for predicting the price.

Applications

- **Predicting prices:** Used to predict the price of a house based on its size, location and other features.
- **Forecasting trends:** Model to forecast the sales of a product based on historical sales data.
- **Identifying risk factors:** Used to identify risk factors for heart patient based on patient medical data.
- **Making decisions:** It could be used to recommend which stock to buy based on market data.

Suggested Quiz 10 Questions

Which of the following regression techniques is specifically designed to handle multicollinearity among independent variables?

A Ridge Regression

B Simple Linear Regression

C Polynomial Regression

D Decision Tree Regression

Login to View Explanation 1/10 < Previous Next >

Comment S Sagar ... + Follow 52

GeeksforGeeks Sanchhaya Education Private Limited

Corporate & Communications Address:
A-143, 7th Floor, Sovereign Corporate Tower, Sector- 136, Noida, Uttar Pradesh (201305)

K 061, Tower K, Gulshan Vivante Apartment, Sector 137, Noida, Gautam Buddh Nagar, Uttar Pradesh, 201305

Company: About Us, Legal, Privacy Policy, Careers, Corporate Solution
Explore: POTD, Practice Problems, Connect, Blogs, Refund on Courses
Tutorials: Programming Languages, DSA, Web Technology, Data Science, DevOps, CS Core Subjects
Courses: ML and Data Science, DSA and Placements, Web Development, Data Science Programming Languages
Offline Centers: Noida, Bengaluru, Pune, Hyderabad
Preparation Corner: Interview Corner, Aptitude, Reasoning, System Design

Python for Machine Learning | Machine Learning with R | Machine Learning Algorithms | EDA | Math for Machine Lea

Sign In

GET IT ON Google Play
Download on the App Store
Training Program
GATE School Subjects
Software and Tools
DevOps & Cloud
GATE
Trending Technologies

@GeeksforGeeks, Sanchhaya Education Private Limited, All rights reserved